

[DASHBOARD](#) / [I MIEI CORSI](#) / [APPELLI DI CLAUDIO SARTORI](#) / [SEZIONI](#) / [MACHINE LEARNING](#) / [MACHINE LEARNING THEORY](#)

Iniziato	Thursday, 13 January 2022, 15:13
Stato	Completato
Terminato	Thursday, 13 January 2022, 15:30
Tempo impiegato	17 min. 13 secondi
Punteggio	15,00/15,00
Valutazione	30,00 su un massimo di 30,00 (100%)

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which is the main reason for the *MinMax scaling* (also known as "*rescaling*") of attributes?

Scegli un'alternativa:

- ☒ Map all the numeric attributes to the same range, in order to prevent the attributes with higher range from having prevalent influence
- ☒ Map all the nominal attributes to the same range, in order to prevent the values with higher frequency from having prevailing influence
- ☐ Remove abnormal values
- ☐ Change the distribution of the numeric attributes, in order to obtain gaussian distributions



Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Given the two binary vectors below, which is their similarity according to the Jaccard Coefficient?

abcde fghij

1000101101

1011101010



Scegli un'alternativa:

- ☐ a. 0.1
- ☐ b. 0.5
- ☐ c. 0.2

☒ ~~x~~ 0.375Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What is the *single linkage*?

Scegli un'alternativa:

- ☐ a. A method to compute the separation of the objects inside a cluster
- ☐ b. A method to compute the distance between two objects, it can be used in hierarchical clustering
- ☒ ~~x~~ A method to compute the distance between two sets of items, it can be used in hierarchical clustering
- ☐ d. A method to compute the distance between two classes, it can be used in decision trees

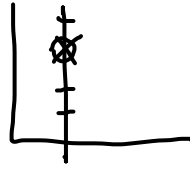
Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Given the definitions below:

- TP = True Positives
- TN = True Negatives
- FP = False Positives
- FN = False Negatives



which of the formulas below computes the *precision* of a binary classifier?

Scegli un'alternativa:

a. $TN / (TN + FP)$

☒ ~~b.~~ $TP / (TP + FP)$

c. $(TP + TN) / (TP + FP + TN + FN)$

~~d.~~ $TP / (TP + FN)$

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Why do we *prune* a decision tree?

Scegli un'alternativa:

~~a.~~ To eliminate rows of the dataset which could be influenced by random effects

☒ ~~b.~~ To eliminate parts of the tree where the decisions could be influenced by random effects

~~c.~~ To eliminate parts of the tree where the decision could generate *underfitting*

~~d.~~ To eliminate attributes which could be influenced by random effects



Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a decision tree, an attribute which is used only in nodes near the leaves...

Scegli un'alternativa:

- ☒ a. ...gives little insight with respect to the target
- ☐ b. ...is irrelevant with respect to the target
- ☐ c. ...has a high correlation with respect to the target
- ☐ d. ...guarantees high increment of purity

Domanda **7**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What are the hyperparameters of a Neural Network? (Possibly non exhaustive)

- ☐ a. Input layers structure, Learning rate, Activation function, Number of epochs
- ☒ b. Hidden layers structure, Learning rate, Activation function, Number of epochs
- ☐ c. Network structure, Learning rate, Backpropagation algorithm, Number of epochs
- ☐ d. Hidden layers structure, Output layer structure, Activation function, Number of epochs



Domanda 8

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following is a strength of the clustering algorithm DBSCAN?

Scegli una o più alternative:

- a. Very fast computation
- ☒ ☒ Ability to find cluster with concavities
- ☒ ☒ Ability to separate outliers from regular data
- d. Requires to set the number of clusters as a parameter

Domanda 9

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What does K-means try to minimise?

Scegli un'alternativa:

- a. The *separation*, that is the sum of the squared distances of each point with respect to its centroid
- b. The *distortion*, that is the sum of the squared distances of each point with respect to the points of the other clusters
- ☒ ☒ The *distortion*, that is the sum of the squared distances of each point with respect to its centroid
- d. The *separation*, that is the sum of the squared distances of each cluster centroid with respect to the global centroid of the dataset



Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the statements below is true? (One or more)

Scegli una o più alternative:

- ☒ ☒ Increasing the radius of the neighbourhood can decrease the number of noise points
- ☐ b. DBSCAN always stops to a configuration which gives the optimal number of clusters
- ☒ ☒ DBSCAN can give good performance when clusters have concavities
- ☒ ☒ Sometimes DBSCAN stops to a configuration which does not include any cluster



Domanda **11**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Match the rule evaluation formulas with their names

$$\frac{\text{conf}(A \Rightarrow C)}{\text{sup}(C)}$$

lift ✓

$$\frac{\text{sup}(A \Rightarrow C)}{\text{sup}(A)}$$

confidence ✓

$$\text{sup}(A \cup C) - \text{sup}(A)\text{sup}(C)$$

leverage ✓

$$\frac{1 - \text{sup}(C)}{1 - \text{conf}(A \Rightarrow C)}$$

altro ✓

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Consider the transactional dataset below

ID Items

- 1 A,B,C
- 2 A,B,D
- 3 B,D,E
- 4 C,D
- 5 A,C,D,E

Which is the *support* of the rule $A, C \Rightarrow B$?

Scegli un'alternativa:

1/5

- ☐ a. 40%
- ☐ b. 100%
- ☒ c. 20%
- ☐ d. 50%



Domanda **13**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following **is not** an objective of feature selection

Scegli un'alternativa:

- a. Avoid the *curse of dimensionality*
- b. Reduce time and memory complexity of the learning algorithms
- ☒ ~~X~~ c. Select the features with higher range, which have more influence on the computations
- d. Reduce the effect of noise

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What is the difference between classification and regression?

- a. Classification is a supervised activity, while regression is unsupervised
- ☒ ~~X~~ b. Classification has a categorical target, while regression has a numeric target
- c. Classification can have a numeric or categorical target, while regression has always a categorical target
- d. Classification can make errors, while regression is always exact



Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What does K-means try to minimise?

Scegli un'alternativa:

- The *distortion*, that is the sum of the squared distances of each point with respect to its centroid
- b. The *distortion*, that is the sum of the squared distances of each point with respect to the points of the other clusters
- c. The *separation*, that is the sum of the squared distances of each point with respect to its centroid
- d. The *separation*, that is the sum of the squared distances of each cluster centroid with respect to the global centroid of the dataset





Vai a...

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